

WEEK 3

Introduction to Proofs

In this lesson, you will be introduced to the different methods used to prove mathematical results.

Objectives

By the end of this week, you should be able to:

- Interpret mathematical proofs and the reasons behind them.
- Prove theorems in Elementary Number Theory using a variety of techniques.

Introduction to Proofs

Proofs in mathematics are valid arguments that establish the truth of mathematical statements.

Methods of Proving: We will learn the following methods of proving theorems: Direct proof, Proof by contraposition, Proof by Contradiction, Vacuous and trivial proofs, Exhaustive Proof, Proof by Cases, Existence Proof.

The proofs are important because they can be used in verifying that computer programs, ensuring that the operating systems are secured, making inferences in artificial intelligence, and showing that system specifications are consistent and so on.

Understanding Numbers: Quick Review

Even Integers: An integer n is even if there exists an integer k such that $n = 2k$.

Odd Integers: An integer n is odd if there exists an integer k such that $n = 2k + 1$ or $n = 2k - 1$.

Rational Numbers: A real number r is rational if there exist integers a and b (with $b \neq 0$) such that $r = a/b$.

Irrational Numbers: A real number that is not rational is called irrational.

Abbreviations

$\mathbb{N}$: Set of natural numbers

$\mathbb{Z}$: Set of integers.

$\mathbb{Q}$: Set of rational numbers.

$\mathbb{Q}'$: Set of irrational numbers.

$\mathbb{R}$: Set of real numbers

Theorem, Proposition, Proof, Axioms, Lemma, Corollary, Conjecture

A theorem is a 'important' statement that can be shown to be true.

'Less important' theorems sometimes are called propositions.

We demonstrate that a theorem is true with a proof: a valid argument that establishes the truth of the theorem.

Statements in a proof can include axioms (or postulates): statements we assume to be true.

A lemma is a 'less important' theorem that is helpful in the proof of other results.

A corollary is a statement that can be established directly from a theorem that has been proven.

A conjecture that is being proposed to be a true statement, usually on the basis on some partial evidence.

3.1 Direct Proofs

Consider the statement: "If p , then q ". In direct proof,

- we assume that p is true;
- then use axioms, definitions, etc to show that q is also true.

Example 3.1. Give a direct proof of the statement: “If n is an odd integer, then n^2 is odd”.

Solution: Assume that n is an odd integer. By definition $n = 2k + 1$ for some integer k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Note that $2k^2 + 2k$ is an integer since k is an integer. Thus by definition, n^2 is odd.

Example 3.2. Prove that the sum of two odd numbers is an even number.

Solution: Let m and n be two odd numbers. By definition of odd numbers:

$$m = 2k_1 + 1 \quad \text{and} \quad n = 2k_2 + 1$$

for some integers k_1 and k_2 . Now

$$m + n = (2k_1 + 1) + (2k_2 + 1) = 2k_1 + 2k_2 + 2 = 2(k_1 + k_2 + 1).$$

Since k_1 and k_2 are integers, it follows that $k_1 + k_2 + 1$ is also an integer. Thus by definition, $m + n$ is an even number.

Exercise:

1. Prove that if m is an even integer and n is an odd integer, then $m + n$ is odd.
2. Give a direct proof of the statement: “If x is a rational number, then x^2 is also rational”.

3.2 Indirect Proofs

Sometimes it may not be possible to prove theorem by using direct proofs.

Eg: Consider the statement: If n is an integer such that n^2 is odd, then n is also odd.

Using direct proof, we first assume that n^2 is an odd integer. So by definition, $n^2 = 2k + 1$ for some integer k . Then $n = \pm\sqrt{2k + 1}$. At this stage, we can not conclude that n is odd. Thus the direct proof fails.

3.2.1 Proof by Contraposition

An important type of indirect proof is Proof by Contraposition. This is based on the fact that for propositions p and q , the statements $p \rightarrow q$ and $\neg q \rightarrow \neg p$ is logically equivalent.

Consider the statement: "If p , then q ". In proof by contraposition, we show that $\neg q \rightarrow \neg p$ is true.

- We assume that $\neg q$ is true;
- then use axioms, definitions, etc to show that $\neg p$ is also true.

Example 3.3. Use contraposition to prove the following:

1. If n is an integer such that $3n + 2$ is even, then n is even.
2. If x and y are real numbers such that $x + y \geq 2$, then either $x \geq 1$ or $y \geq 1$.

Solution

1. Assume that n is not an even integer. So n must be odd. By definition

$$n = 2k + 1 \text{ for some integer } k.$$

Then

$$\begin{aligned} 3n + 2 &= 3(2k + 1) + 2 \\ &= 6k + 5 \\ &= 2(3k + 2) + 1 \end{aligned}$$

Note that $3k + 2$ is an integer since k is an integer. Thus by definition, $3n + 2$ is odd. We have shown that if n is not even, then $3n + 2$ is not even. In other words, if $3n + 2$ is an even integer, then n is even.

2. Assume that $x < 1$ and $y < 1$. Then

$$x + y < 1 + 1 = 2.$$

Thus we have shown that if $x < 1$ and $y < 1$, then $x + y < 2$. In other words, if $x + y \geq 2$, then either $x \geq 1$ or $y \geq 1$.

Exercise: Use contraposition to prove that if x is irrational, then $1/x$ is irrational.

3.2.2 Proof by Contradiction

In proof by contradiction, we show that the negation of the given statement is false.

- We assume that the negation is true;

- then use axioms, definitions, etc to show that this leads to a contradiction.

(Note: $\neg(\forall x(p \rightarrow q)) \equiv \exists x(p \wedge \neg q)$)

Example 3.4. Use proof by contradiction to show that

1. If n is an integer such that n^2 is odd, then n is also odd.
2. If x is irrational, then $1/x$ is irrational.

Solution

1. Assume that n^2 is an odd integer and n is not odd. If n is not odd, then it must be even. By definition,

$$n = 2k \text{ for some integer } k.$$

Then

$$\begin{aligned} n^2 &= (2k)^2 \\ &= 4k^2 \\ &= 2(2k^2) \end{aligned}$$

This shows that n^2 is even which contradicts our assumption that n^2 is odd. Thus if n^2 is an odd integer, then n is also odd.

2. Assume that x is irrational and $1/x$ is rational. By definition

$$\frac{1}{x} = \frac{a}{b} \text{ where } a \text{ and } b \text{ are integers with } b \neq 0.$$

Since $1/x$ can not be zero (why?), we know that $a \neq 0$. Now

$$x = \frac{1}{1/x} = \frac{1}{a/b} = \frac{b}{a}.$$

This shows that x is rational which contradicts our assumption that x is irrational. Thus if x is irrational, then $1/x$ is irrational.

3.3 Vacuous Proofs

- From the truth table of $p \rightarrow q$, we see that when p is false, $p \rightarrow q$ is true.
- Hence if we can show that the antecedent (p) is false, then it means that we have proved that $p \rightarrow q$ is true. This is called vacuous proof.

Example 3.5. Show that the proposition $P(0)$ is true where $P(n)$ is: "If $n > 1$, then $n^2 > n$ ".

Solution: Note that $P(0)$ is "If $0 > 1$, then $0^2 > 0$ ". Since the antecedent ($0 > 1$) is false, $P(0)$ is true.

3.4 Trivial Proofs

- From the truth table of $p \rightarrow q$, we see that when q is true, $p \rightarrow q$ is also true.
- Hence if we can show that the conclusion (q) is true, then it means that we have proved that $p \rightarrow q$ is true. This is called trivial proof.

Example 3.6. Show that the proposition $P(0)$ is true, where $P(n)$ is: If a and b are positive integers with $a \geq b$, then $a^n \geq b^n$.

Solution: Note that $P(0)$ is "If $a \geq b$, then $a^0 \geq b^0$ ". Since the conclusion ($a^0 \geq b^0$) is true, $P(0)$ is true.

3.5 Exhaustive Proof

- In this proof, we need to show that the theorem is true for all possible values in the domain.
- This proof is useful when the domain contains a relatively small amount of finite values.

Example 3.7. Prove that $(n + 1)^2 \geq 2^n$ if n is a positive integer with $n < 4$.

Solution: We need to verify that $(n + 1)^2 \geq 2^n$ when $n = 1, 2$ and 3 .

For $n = 1$, we have $(n + 1)^2 = (1 + 1)^2 = 4$ and $2^n = 2^1 = 2$.

For $n = 2$, we have $(n + 1)^2 = (2 + 1)^2 = 9$ and $2^n = 2^2 = 4$.

For $n = 3$, we have $(n + 1)^2 = (3 + 1)^2 = 16$ and $2^n = 2^3 = 8$.

In each of the three cases we see that $(n + 1)^2 \geq 2^n$. Thus $(n + 1)^2 \geq 2^n$ if n is a positive integer with $n < 4$.

3.6 Proof by Cases

A proof by cases must cover all possible cases in a theorem.

Example 3.8. Prove that if n is an integer then $n^2 \geq n$.

Solution: We can prove this by considering three cases; when $n > 0$, when $n < 0$ and when $n = 0$.

Case 1: When $n > 0$: $n > 0$ implies $n \geq 1$. Multiply both sides by n (positive integer), we get $n^2 \geq n$.

Case 2: When $n < 0$: $n < 0$ implies $n \leq -1$. Multiply both sides by n (negative integer), we get $n^2 \geq -n$. However $-n \geq n$ (since n is a negative integer). Thus $n^2 \geq n$.

Case 3: When $n = 0$: $0^2 \geq 0$ is true.
Thus if n is an integer then $n^2 \geq n$.

3.7 Existence Proof

- A proof of the proposition of the form $\exists xP(x)$ is called an existence proof.
- $\exists xP(x)$ can be proved by giving an example that can make $P(x)$ true.

Example 3.9. *Prove that there exist a rational number x and irrational number y such that x^y is rational.*

Solution: We can prove this by giving an example that can make the above statement true: If $x = 1 \in \mathbb{Q}$ and $y = \sqrt{2} \in \mathbb{Q}'$, we see that $x^y = 1^{\sqrt{2}} = 1 \in \mathbb{Q}$. Thus there exist a rational number x and irrational number y such that x^y is rational.

3.8 Disproving a Statement

A statement of the form $\forall xP(x)$ can be disproved by giving a counterexample.

Example 3.10. *Disprove that if a and b are rational number then a^b is also rational.*

Solution: We look for a counterexample. Let $a = 2 \in \mathbb{Q}$ and $b = \frac{1}{2} \in \mathbb{Q}$, we see that $a^b = 2^{\frac{1}{2}} = \sqrt{2} \notin \mathbb{Q}$. Thus we have disproved that if a and b are rational number then a^b is also rational.